

Chemical and hydraulic engineers are often interested in computing the settling velocity of a suspension of particulates in a given liquid. For dilute suspensions, a variable that is directly related with this settling velocity is the so-called terminal velocity, u_t , which is the constant falling velocity of an isolated particle moving in an infinite fluid under the action of gravity. For the particular case of slime sedimentation in a river, the slime particles are spheres of average diameter $D_s = 30\mu m$ and average mass $m_s = 0.2\mu g$, much too small to be handled in the lab. To guide the design of an experiment to determine u_t , the following steps are suggested:

1. Use the Π theorem to reduce the parametric dependence of the terminal velocity.
2. To enable the use of larger particles in the experiment, explore the replacement of water with glycerine ($\nu_g/\nu_w = 680$, $\rho_g/\rho_w = 1.25$) and find the corresponding values of the diameter D_e and mass m_e for the resulting experimental particle.
3. For this particle, the experimental measurements give $u_e = 10^{-3}$ m/s. Obtain the terminal velocity for an average slime particle.

$$\textcircled{1} \quad u_t = u_t(\underline{D}, \underline{m}, \underline{g}, \underline{\nu}, \underline{\rho}) \rightarrow \frac{u_t}{\sqrt{gD}} = f\left(\frac{m}{\rho D^3}, \frac{\nu}{g^{1/2} D^{3/2}}\right)$$

$$\textcircled{2} \quad \text{FOR THE EXPERIMENT TO BE PHYSICALLY SIMILAR TO REALITY:}$$

$$\frac{m_s}{\rho_w D_s^3} = \frac{m_e}{\rho_g D_e^3} \Rightarrow m_e = \frac{\rho_g}{\rho_w} \left(\frac{D_e}{D_s}\right)^3 m_s =$$

$$\frac{\nu_w}{g^{1/2} D_s^{3/2}} = \frac{\nu_g}{g^{1/2} D_e^{3/2}} \Rightarrow D_e = \left(\frac{\nu_g}{\nu_w}\right)^{2/3} D_s = 2.32 \text{ mm}$$

$$\textcircled{3} \quad \frac{u_t}{\sqrt{gD_s}} = \frac{u_e}{\sqrt{gD_e}} \rightarrow u_t = \sqrt{D_s/D_e} u_e = 0.1137 \text{ mm/s}$$